

Post-Quantum Cryptography: The Threats of Quantum Computing to Security

ENGG*4560

04/4/2025

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Abstract—The rapid progress of quantum computing poses a serious threat to classical cryptographic systems, especially public key algorithms like RSA and ECC that secure today’s digital communications. Researchers estimate that within the next 10 to 15 years, post quantum resistant security will become mandatory for cryptographic protocols [1]. In response, post quantum cryptography (PQC) has appeared, designing new public key algorithms based on mathematical problems believed to resist quantum attacks. In parallel, quantum cryptography techniques such as Quantum Key Distribution (QKD) offer information theoretical security based on the laws of physics. This survey explores the fundamental concepts of PQC and quantum computing threats, reviews common design paradigms for achieving quantum safe security including PQC algorithms, QKD systems, and hybrid approaches, and presents case studies illustrating these designs in practice. Two representative examples are discussed: one integrating a lattice based PQC algorithm to augment classical RSA cryptography, and another deploying a QKD network and analyzing its vulnerabilities. The paper concludes that a holistic strategy combining post quantum algorithms with quantum-based methods is essential to protect communications in the quantum era, ensuring long term security and privacy.

Keywords—*RSA; ECC; Quantum Key Distribution; QKD network vulnerabilities; long-term security; privacy protection*

I. INTRODUCTION

Widely used communication protocols and public key cryptosystems are not secure against an adversary with a quantum computer [1]. Protocols like TLS 1.3 and the Signal messaging protocol rely on Diffie Hellman key exchanges that could be broken by a sufficiently powerful quantum attacker [1], [2]. It is expected that post quantum security will become mandatory for new cryptographic standards within the next decade or two [1]. Urgency is amplified by ongoing breakthroughs in quantum hardware; for example, recent improvements in qubit error rates bring large scale quantum computers closer to reality. Once such machines are built, they could recover the private keys of most current asymmetric cryptosystems by exploiting quantum algorithms [1]. In fact, encrypted data intercepted today might be decrypted retroactively by future quantum adversaries, jeopardizing long term confidentiality [4]. To address these threats, the cryptographic community is exploring quantum safe security solutions. Two broad paradigms have gained prominence: post quantum cryptography (PQC) and quantum cryptography. PQC refers to conventional, non-quantum cryptographic algorithms designed to resist attacks by quantum computers [3]. In contrast, quantum cryptography uses quantum physics for security; a notable example is Quantum Key Distribution (QKD), which enables two parties to share encryption keys with unconditional security (in theory) guaranteed by quantum mechanics [3]. Both approaches have distinct strengths and limitations. This survey provides an overview of the fundamental concepts in PQC and quantum computing threats, then examines common design paradigms for quantum resistant

security such as new algorithmic constructions and hybrid systems. Case studies are presented that focus on integrating PQC with classical cryptography and on QKD deployment. Finally, it is concluded that a combined approach integrating PQC with quantum-based methods is likely necessary for secure communications in the quantum era.

II. BACKGROUND

Modern public key cryptography (PKC) relies on mathematical problems that are intractable for classical computers but can be efficiently solved by a quantum computer. A prominent example is the RSA algorithm, whose security is based on the difficulty of factoring large integers. While factoring is sub exponential on classical machines, Shor's quantum algorithm can factor integers in polynomial time [1]. Similarly, the discrete logarithm problem underlying Diffie Hellman and elliptic curve cryptography (ECC) can be solved quickly using Shor's algorithm [3]. This means that RSA, Diffie Hellman, and ECC would be completely broken given a sufficiently powerful quantum computer [1]. In fact, a quantum computer with only a few thousand stable qubits could recover the private keys of most current asymmetric cryptosystems. This impending capability defines the Y2Q (Years to Quantum) problem, which is the countdown until quantum computers can defeat today's encryption. Symmetric key cryptography is more secure against quantum attacks but not immune. Algorithms like AES and SHA 256 do not suffer from the weaknesses exploited by Shor's algorithm; however, Grover's algorithm can quadratically speed up brute force searches, thus reducing the security level by half [3]. In

practice, this means that doubling the key lengths, for example, from 128 bit to 256 bit for AES can counteract Grover's algorithm. Thus, while symmetric cryptography can be made quantum safe with larger parameters, public key cryptography faces a much more severe challenge. A critical vulnerability in many secure communication protocols is the key exchange mechanism [1]. Most current protocols, such as TLS, VPNs, and secure messaging, use Diffie Hellman key exchanges as seen in *Fig. 1*, or ECC based key exchanges [1], [3]. Because these are vulnerable to quantum attacks, an eavesdropper could record encrypted traffic now and later use a quantum computer to derive the session keys and decrypt the content. This "harvest now, decrypt later" attack is the main threat to secure communication and messaging protocols in a post quantum context [1][4].

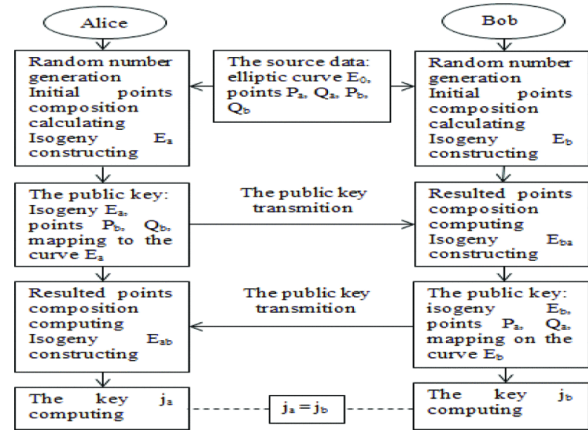


Fig. 1. Key exchange flowchart (source [1]).

Post quantum cryptography refers to new cryptographic algorithms, especially public key algorithms for encryption, key exchange, and digital signatures, which are designed to be resistant to quantum computer attacks [3]. These algorithms, implemented on conventional computers, rely on mathematical problems outside the scope of

Shor's or Grover's algorithms. Several families of hard problems form the foundation of current PQC research [3]. Lattice based cryptography relies on problems such as the Shortest Vector Problem and Learning with Errors (LWE); lattice-based schemes, such as CRYSTALS Kyber for key encapsulation and CRYSTALS Dilithium for signatures, have strong security proofs and efficient implementations, making them leading candidates for standardization [3], [2]. Code based cryptography is based on error correcting code problems like syndrome decoding, as seen in the McEliece encryption scheme [2]. While offering security, these schemes often suffer from large public key sizes. Hash based cryptography uses cryptographic hash functions to build digital signatures (for example, XMSS or SPHINCS+); although these can produce larger signatures and slower verification speeds, their security relies solely on the strength of the underlying hash function [2]. Multivariate quadratic cryptography is based on the difficulty of solving systems of multivariate quadratic equations over finite fields; some schemes in this category have been broken, while others continue to be evaluated. Isogeny based cryptography leverages mathematical structures from elliptic curve isogenies, as in the SIKE key exchange, which features small key sizes, although certain isogeny schemes have been subject to new cryptanalytic attacks. Each of these families has tradeoffs in terms of key sizes, computational efficiency, and security margins. In general, PQC algorithms tend to have larger public keys and higher computational costs compared to classical systems. To ease adoption, the US National Institute of Standards and Technology (NIST) started an open competition to

standardize post quantum algorithms, with several finalists appearing as strong candidates for future deployment.

In contrast to the mathematical approach of PQC, quantum cryptography uses the laws of physics for security. The most well-known application is Quantum Key Distribution (QKD), which enables two parties to generate a shared secret key over a quantum channel with security based on quantum mechanics. The principle behind QKD is that measuring a quantum state disturbs it, so an eavesdropper trying to intercept the quantum signals will introduce detectable errors. After the quantum transmission, the legitimate parties use an authenticated classical channel to perform error correction and privacy amplification, yielding a shared secret key that is theoretically secure [1], [3]. While QKD provides unconditional security, it requires specialized hardware, such as single photon sources and detectors, and suffers from practical constraints including distance limitations and the need for trusted relay nodes. Moreover, the security of QKD depends on both the quantum channel and the classical post processing procedures. Implementation vulnerabilities, such as detector side channel attacks, can compromise the security of a QKD system.

III. DESIGN PARADIGMS FOR QUANTUM RESISTANT SECURITY

Post-Quantum Algorithm Type	Third Round Finalist	Technology	Alternate Candidates	Technology
Public Key Encryption/Key Encapsulation Mechanisms	Classic McEliece	Code	BIKE	Code
	CRYSTALS KYBER	Lattice	ProdoKEM	Lattice
	NTRU	Lattice	HQC	Code
	SABER	Lattice	SIKE	Supersingular Isogeny
Digital Signature Algorithms	CRYSTALS-DILITHIUM	Lattice	GeMSS	Multivariate Polynomial
	FALCON	Lattice	PICNIC	Other
	RAINBOW	Multivariate Polynomial	SPHINCS+	Hash

Fig. 2. NIST Round 3 finalists for PQC algorithms' standardization (source [4]).

A primary design paradigm is the creation of new cryptographic algorithms that can replace classical public key systems. PQC algorithms are built upon mathematical problems believed to be hard for quantum computers. For example, lattice-based schemes rely on problems for which no efficient quantum algorithm is known. The NIST PQC competition is an open evaluation process designed to ensure that new PQC algorithms meet stringent security and performance criteria. *Fig. 2.* illustrates the outcome of the NIST post-quantum cryptography project's third round, highlighting the finalist algorithms in line for standardization. In many cases, designers aim for algorithms that are as compatible as possible with existing protocols so that traditional systems, such as TLS, can be updated with new cipher suites.

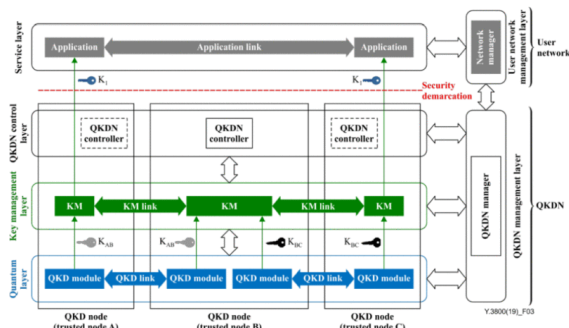


Fig. 3. The six-layered, modular structure of a QKD Network (source [4]).

Another design paradigm focuses on quantum based cryptographic systems like QKD [4]. As shown in *Fig. 3.*, a QKD network can be organized into a six-layer architecture, from the quantum physical layer up to the key management layer. This layered model (standardized by ITU-T) illustrates how quantum keys are distributed across different network levels. QKD system design must encompass not only the quantum

protocol but also the supporting hardware and network architecture. Standards are being developed to address vulnerabilities in QKD implementations, including those related to hardware imperfections and post processing. For example, QKD networks typically involve a layered architecture with dedicated quantum channels and classical channels for error correction and authentication [3]. System design in this area also involves defense in depth strategies, where multiple security measures are integrated to address both quantum and classical threats. Furthermore, Recent research has led to improved designs, such as measurement-device-independent QKD (MDI-QKD), which mitigates detector vulnerabilities by removing the need to trust the measurement devices [4]. Additionally, integrating QKD into network architecture further requires developing standards and protocols for key relaying and integration with classical networks.

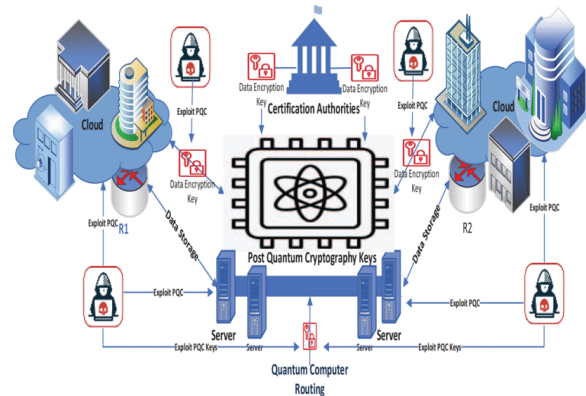


Fig. 4. PQC Key Exchanges Systems and Vulnerable Spots (source [2]).

IV. PQC KEY EXCHANGES SYSTEMS AND VULNERABLE AREAS

A popular approach is the hybrid design that combines classical cryptographic techniques

with PQC or QKD methods as seen in *Fig. 4*. Hybrid architectures aim to use the strengths of both worlds while compensating for their individual weaknesses [2]. For instance, a TLS handshake might use both a classical key exchange, such as ECDH, and a post quantum key exchange, such as a lattice based KEM, ensuring that the session stays secure even if one method is later compromised [1]. Similarly, QKD can be used alongside post quantum cryptography to protect the key exchange process while also relying on well-established symmetric algorithms for data encryption [2]. This layered approach provides redundancy and resilience during the transition to fully quantum safe systems.

V. CASE STUDIES

An example of post quantum deployment is the integration of a lattice based key encapsulation mechanism (KEM) alongside classical RSA encryption in a client server scenario. In this hybrid RSA plus PQC scheme, a standard RSA 2048 encryption is combined with a lattice-based algorithm, such as CRYSTALS Kyber, to encrypt the same message. An attacker must break both algorithms to compromise the communication. Experimental results have shown that lattice-based schemes offer significantly improved performance in key generation and encryption compared to RSA, while simultaneously resisting quantum attacks. This hybrid approach provides immediate security benefits without requiring a complete overhaul of existing public key infrastructures.

Another example is the deployment of a QKD network in a metropolitan area. QKD

enables the generation of shared secret keys with theoretical unconditional security, but practical implementations require an authenticated classical channel and error correction. In one scenario, two network nodes use the BB84 protocol to share a key. However, if the classical authentication mechanism relies on vulnerable RSA or ECC signatures, a quantum adversary could potentially compromise the system. Moreover, hardware vulnerabilities, such as detector side channel attacks, pose added risks. To mitigate these issues, hybrid designs are proposed in which QKD is combined with post quantum digital signatures for authentication, ensuring that both the quantum and classical components are resistant to quantum era attacks. This case study highlights the importance of comprehensive system design that addresses vulnerabilities across the entire network.

CONCLUSION

Quantum computing on one hand threatens to break the cryptographic foundations that protect digital communications, particularly public key algorithms for key exchange and digital signatures [4]. On the other hand, it has spurred the development of post quantum cryptography and quantum cryptographic methods such as QKD. This survey has reviewed the quantum computing threats to classical cryptography, examined the fundamental concepts behind PQC and QKD, and presented key design paradigms for achieving quantum resistant security. Two case studies proved the practical benefits of integrating lattice based PQC with classical RSA systems and the challenges inherent in QKD networks. The overall conclusion is that a multi layered, hybrid approach combining post quantum algorithms with

quantum-based techniques is essential to secure communications in the quantum era. Continued research, and practical testing are critical to ensure that our digital infrastructure holds against emerging quantum threats.

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